Birzeit University

Discrete Mathematics (COMP 233): First Exam

Section 9: Answers & Marking Scheme

Second Semester 2025/2026

Question One [10 Marks]: Multiple-choice (2 marks each, no partial credit)

Marking key, Question One (quick reference)

Q	Answer	Marks	Topic / what to check
1.1	**(c) 4**	2	Quotient–remainder: square of $n = 5k + 2$ mod 5.
1.2	**(b) 6^5**	2	Prime factorisation: highest power of 6 dividing $30^5 \cdot 40^3$.
1.3	**(b) 1**	2	Modular arithmetic: reducing $6(a+b) + 7$ mod 6.
1.4	**(d)**	2	Universal vs existential: distinguishing ∀ from ∃ restatements.
1.5	**(b)**	2	Existential over reals: $x^2 + 1 = 0$ has no real solution.
Total		**10**	**Mark the selected option only; do not award partial credit.**

Q1.1 [2 marks]

> If $n = 5k + 2$ for some integer k, what is $n^2 \mod 5$?

ANSWER (c) 4.

SOLUTION $n^2 = (5k + 2)^2 = 25k^2 + 20k + 4 = 5(5k^2 + 4k) + 4$. The first term is a multiple of 5; the leftover is 4. Hence $n^2 \mod 5 = 4$.

Q1.2 [2 marks]

> What is the largest power of 6 that divides $30^5 \cdot 40^3$?

ANSWER (b) 6^5.

SOLUTION Factorise each base: $30 = 2 \cdot 3 \cdot 5$ gives $30^5 = 2^5 \cdot 3^5 \cdot 5^5$; $40 = 2^3 \cdot 5$ gives $40^3 = 2^9 \cdot 5^3$. The product is $2^{14} \cdot 3^5 \cdot 5^8$. Each factor of $6 = 2 \cdot 3$ consumes one 2 and one 3; the limiting count is $min(14, 5) = 5$. Largest power of 6: 6^5.

Q1.3 [2 marks]

> If m and n are integers with $m \bmod 6 = 3$ and $n \bmod 6 = 4$, what is $(m + n) \bmod 6$?

ANSWER (b) 1.

SOLUTION Write $m = 6a + 3$ and $n = 6b + 4$. Then $m + n = 6(a + b) + 7 = 6(a + b + 1) + 1$. Because a remainder must lie in $[0, 6)$, $(m + n) \bmod 6 = 1$. *Common trap:* option (a) "7" results from stopping at $6(a+b) + 7$ without reducing.

Q1.4 [2 marks]

> Which of (a)–(d) is NOT equivalent to "$\forall$ *integers n, if 4 | n then 2 | n*"?

ANSWER (d).

SOLUTION Options (a), (b), and (c) are standard universal restatements of the same conditional. Option (d) is existential ("some integer..."), which is logically weaker than a universal claim and is therefore not equivalent.

Q1.5 [2 marks]

> Consider "$\exists$ *a real number x such that* $x^2 + 1 = 0$". Which best describes it?

ANSWER (b).

SOLUTION For every real x, $x^2 \geq 0$, hence $x^2 + 1 \geq 1 > 0$. No real solution exists, so the existential claim is false. Options (a) and (c) test simple substitutions ($0^2 + 1 = 1$; $(-1)^2 + 1 = 2$). The complex roots $\pm i$ are not real numbers, so (d) misreads the domain.

Question Two [10 Marks]: Direct Proofs

Q2.A [4 marks]

> Prove that for every odd integer n, $n^2 + 3$ is divisible by 4.

SOLUTION Let n be any odd integer. We must show that $4 \mid (n^2 + 3)$.

By the definition of odd, $n = 2k + 1$ for some integer k.

Then $n^2 + 3 = (2k + 1)^2 + 3 = 4k^2 + 4k + 1 + 3 = 4k^2 + 4k + 4 = 4(k^2 + k + 1)$.

Let $m = k^2 + k + 1$. Then m is an integer (integers are closed under products and sums). So $n^2 + 3 = 4m$, and by the definition of divisibility, $4 \mid (n^2 + 3)$.

Marking, Q2.A [4 marks]

Marks	Criterion
1	Correct setup using the definition of odd: $n = 2k + 1$ for some integer k.
1	Correct algebraic expansion: $(2k + 1)^2 + 3 = 4k^2 + 4k + 4$.
1	Factoring out 4 to obtain $4(k^2 + k + 1)$.
1	Clean conclusion: name the integer factor and cite the definition of divisibility.

Q2.B [3 marks]

> Prove that for all integers a, b, c: if $a \mid b$ and $b \mid c$ then $a \mid c$.

SOLUTION Let a, b, c be integers with $a \mid b$ and $b \mid c$. We must show $a \mid c$.

By the definition of divisibility, there exist integers r and s with $b = ar$ and $c = bs$.

Substituting: $c = bs = (ar)s = a(rs)$. Let $k = rs$; k is an integer (closure under product). Hence $c = ak$, and by the definition of divisibility, $a \mid c$.

Marking, Q2.B [3 marks]

Marks	Criterion
1	Correct application of the definition of divisibility to both hypotheses: $b = ar$ and $c = bs$.
1	Correct substitution to express $c = a(rs)$.
1	Clean conclusion: rs is an integer, hence $a \mid c$.

Q2.C [3 marks]

> Use the quotient–remainder theorem with $d = 3$ to show that for every integer n, $n^2 + 2n$ has the form $3k$ or $3k + 2$ for some integer k.

SOLUTION Let n be any integer. By the quotient–remainder theorem with $d = 3$, exactly one of the following holds for some integer q:

$$n = 3q, \quad n = 3q + 1, \quad \text{or} \quad n = 3q + 2.$$

Case 1 ($n = 3q$): $n^2 + 2n = 9q^2 + 6q = 3(3q^2 + 2q)$. Form $3k$, with $k = 3q^2 + 2q$.

Case 2 ($n = 3q + 1$): $n^2 + 2n = (3q + 1)^2 + 2(3q + 1) = 9q^2 + 12q + 3 = 3(3q^2 + 4q + 1)$. Form $3k$.

Case 3 ($n = 3q + 2$): $n^2 + 2n = (3q + 2)^2 + 2(3q + 2) = 9q^2 + 18q + 8 = 3(3q^2 + 6q + 2) + 2$. Form $3k + 2$.

In all three cases, $n^2 + 2n$ has the form $3k$ or $3k + 2$.

Marking, Q2.C [3 marks]

Marks	Criterion
1	Correct setup applying the quotient–remainder theorem to split n into three cases.
1	Correct algebra for $n^2 + 2n$ in all three cases.
1	Correct identification of the target form ($3k$ or $3k + 2$) for each case and clean overall conclusion.

Question Three [10 Marks]: Counterexample & Closure

Q3.A [4 marks]

> Disprove "for all integers m and n, if both are odd then $m + n$ is odd"; state the correct conclusion and prove it.

COUNTEREXAMPLE Take $m = 1$ and $n = 3$. Both are odd, but $m + n = 4$, which is even. The statement is false.

Acceptable variants: any specific pair of odd integers whose sum is even, e.g. $(1, 1)$, $(3, 5)$, $(-1, 7)$.

CORRECTED STATEMENT For all integers m and n, if m and n are both odd, then $m + n$ is even.

PROOF Let m and n be integers, both odd. By the definition of odd, $m = 2j + 1$ and $n = 2k + 1$ for some integers j and k.

Then $m + n = (2j + 1) + (2k + 1) = 2j + 2k + 2 = 2(j + k + 1)$. Let $q = j + k + 1$; q is an integer. So $m + n = 2q$, and by the definition of even, $m + n$ is even.

Marking, Q3.A [4 marks]

Marks	Criterion
1	Valid counterexample: any specific pair of odd integers whose sum is even (no partial credit for a generic claim without a witness).
1	Correct statement of the revised conclusion ($m + n$ is even when both are odd).
1	Correct setup using the definition of odd: $m = 2j + 1$, $n = 2k + 1$.
1	Completing the proof: show $m + n = 2 \cdot (integer)$ and conclude even.

Q3.B [6 marks]

> Prove that for all rational numbers r and s, the number $r^2 + s^2$ is rational.

SOLUTION Let r and s be rational. We must show $r^2 + s^2$ is rational.

By the definition of rational, there exist integers a, b, c, d with $b \neq 0$ and $d \neq 0$ such that $r = a/b$ and $s = c/d$.

Then $r^2 = a^2/b^2$ and $s^2 = c^2/d^2$. Combining over the common denominator b^2d^2:

$$r^2 + s^2 = a^2/b^2 + c^2/d^2 = (a^2d^2 + b^2c^2) / (b^2d^2).$$

The numerator $a^2d^2 + b^2c^2$ is an integer (closure of integers under product and sum).

The denominator b^2d^2 is an integer, and it is nonzero: since $b \neq 0$ and $d \neq 0$, neither b^2 nor d^2 is zero, so their product is nonzero.

Hence $r^2 + s^2$ is the quotient of an integer by a nonzero integer; by the definition of rational, it is rational.

Marking, Q3.B [6 marks]

Marks	Criterion
1	Correct setup using the definition of rational: $r = a/b$, $s = c/d$, with $b, d \neq 0$.
1	Correct squaring: $r^2 = a^2/b^2$ and $s^2 = c^2/d^2$.
1	Correct combination into a single fraction with common denominator b^2d^2.
1	Justifying that the numerator $a^2d^2 + b^2c^2$ is an integer.
1	Justifying that the denominator b^2d^2 is a nonzero integer (frequently missed).
1	Clean conclusion citing the definition of rational.

Question Four [10 Marks]: Predicate Logic & Validity

Q4.A.a [3 marks]

> Let S be the set of students, C the set of courses, and $E(s, c)$ the predicate "student s is enrolled in course c". Write a symbolic form of: *"there is a course in which every student is enrolled."*

ANSWER $\exists\, c \in C$ such that $\forall\, s \in S$, $E(s, c)$.

NOTE Structure is $\exists\forall$: the same course c must enrol every student s. *Common error:* writing $\forall\, s$, $\exists\, c$, $E(s, c)$ says every student is enrolled in some course (possibly different for each student), which is strictly weaker and not equivalent.

Marking, Q4.A.a [3 marks]

Marks	Criterion
1	Correct outer quantifier $\exists\, c \in C$.
1	Correct inner quantifier $\forall\, s \in S$, with proper scope.
1	Correct predicate $E(s, c)$ with the arguments in the right order.

Q4.A.b [3 marks]

> Write the negation of $\forall\, s \in S$, $\exists\, c \in C$ such that $E(s, c)$.

ANSWER $\exists\, s \in S$ such that $\forall\, c \in C$, $\sim E(s, c)$.

WORKING $\sim(\forall\, s, \exists\, c, E(s, c)) \equiv \exists\, s, \sim(\exists\, c, E(s, c)) \equiv \exists\, s$ such that $\forall\, c, \sim E(s, c)$.

ENGLISH "There is a student who is not enrolled in any course."

Marking, Q4.A.b [3 marks]

Marks	Criterion
1	Correct conversion of the outer $\forall$ to $\exists$.
1	Correct conversion of the inner $\exists$ to $\forall$.
1	Correct negation of the predicate: $E(s, c)$ becomes $\sim E(s, c)$.

Q4.B [4 marks]

> Determine the validity of the argument with premises $p \vee q$, $p \to r$, $q \to r$ and conclusion $\therefore r$.

SOLUTION Build the truth table (T = true, F = false). Premise columns are $p \vee q$, $p \to r$, $q \to r$; the conclusion column is r.

p	q	r	p ∨ q	p → r	q → r	r	
T	**T**	**T**	**T**	**T**	**T**	**T**	**critical**
T	T	F	T	F	F	F	
T	**F**	**T**	**T**	**T**	**T**	**T**	**critical**
T	F	F	T	F	T	F	
F	**T**	**T**	**T**	**T**	**T**	**T**	**critical**
F	T	F	T	T	F	F	
F	F	T	F	T	T	T	
F	F	F	F	T	T	F	

VERDICT The argument is **valid**. There are three critical rows where all three premises are true: *(p, q, r) = (T, T, T), (T, F, T), (F, T, T)*. In every critical row the conclusion r is also true, so there is no row where all premises are true but the conclusion is false. (This is the standard proof-by-cases inference rule.)

Marking, Q4.B [4 marks]

Marks	Criterion
1	Complete and correct 8-row truth table with all premise and conclusion columns computed correctly.
1	Correct identification of the premise columns ($p \vee q$, $p \to r$, $q \to r$) and the conclusion column (r).
1	Correct identification of the three critical rows where all premises are true.
1	Correct validity verdict with reasoning that references the critical rows.

Marks summary

Question One	5 MCQs × 2 marks	**10**
Question Two	4 + 3 + 3	**10**
Question Three	4 + 6	**10**
Question Four	(3 + 3) + 4	**10**
Total		**40**

General Marking Principles

- For MCQs (Q1), award 2 marks for the correct selected option and 0 otherwise. No partial credit.
- For proof questions, the marking breakdown is sequential: students lose individual marks for missing or incorrect steps, not the whole question.
- Award full credit when the student uses different variable names (e.g. j, k, p, q) provided the logic is correct.
- Accept any logically equivalent reformulation of the symbolic answers in Q4.A.
- In Q3.A, any specific pair of odd integers with even sum is acceptable. No partial credit for a generic claim without a specific witness.
- In Q4.B, accept alternative correct column orderings provided premise and conclusion columns are clearly identified.